Research Notes : Index Density

In analytically evalauting $\rho(z)$,

$$
\begin{aligned}
\rho_{\text{ind}}(z) =_{(1)}\ & \pi^{-2(n+1)} \int \mathrm{d}^2X\ \mathrm{d}^{2n}\phi\ e^{-|X|^2-|\phi|^2}|X|^2 \det \begin{pmatrix} \delta^I{}_J\overline{X} - \overline{\phi}^I\phi_J/X & \mathcal{F}_{IJK}\overline{\phi}^K \\ \overline{\mathcal{F}}^{IJK}\phi_K & \delta_I{}^J X - \phi_I\overline{\phi}^J/\overline{X} \end{pmatrix} \\
=_{(2)}\ & \pi^{-2(n+1)} \int \mathrm{d}^2X\ \mathrm{d}^{2n}\phi\ e^{-|X|^2-|\phi|^2}|X|^2 \det\Big(\delta^I{}_J\overline{X} - \overline{\phi}^I\phi_J/X\Big) \det\Big(\delta_I{}^J X - \phi_I\overline{\phi}^J/\overline{X}\Big) \\
& \qquad \times \det\Big(\delta_I{}^P - |X|^{-2}(\mathcal{F}_{IJK}\overline{\phi}^K)(\delta_J{}^L - \phi_J\overline{\phi}^L/|X|^2)^{-1}(\overline{\mathcal{F}}^{LMN}\phi_N)(\delta^M{}_P - \overline{\phi}^M\phi_P/|X|^2)^{-1}\Big) \\
=_{(3)}\ & \pi^{-2(n+1)} \int \mathrm{d}^2X\ \mathrm{d}^{2n}\phi\ e^{-|X|^2-|\phi|^2}|X|^{2n-2}(|X|^2-|\phi|^2)^2 \\
& \qquad \times \det\left(\delta_I{}^P - |X|^{-2}(\mathcal{F}_{IJK}\overline{\phi}^K)\left(\delta^J{}_L + \frac{\overline{\phi}^J\phi_L}{|X|^2-|\phi|^2}\right)(\overline{\mathcal{F}}^{LMN}\phi_N)\left(\delta_M{}^P + \frac{\phi_M\overline{\phi}^P}{|X|^2-|\phi|^2}\right)\right) \\
=_{(4)}\ & \pi^{-2(n+1)} \int \mathrm{d}^2X\ \mathrm{d}^{2n}\phi\ e^{-|X|^2-|\phi|^2}|X|^{2n-2}D^4 \det\Big(\delta_I{}^P - \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JPN}\overline{\phi}^K\phi_N|X|^{-2} - \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}\overline{\phi}^K\overline{\phi}^P\phi_M\phi_N|X|^{-2}/D^2 \\
& \qquad - \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LPN}\overline{\phi}^J\overline{\phi}^K\phi_L\phi_N|X|^{-2}/D^2 - \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}\overline{\phi}^J\overline{\phi}^K\overline{\phi}^P\phi_L\phi_M\phi_N|X|^{-2}/D^4\Big) \\
=_{(5)}\ & \pi^{-2(n+1)} \int \mathrm{d}^2X\ \mathrm{d}^{2n}\phi\ e^{-|X|^2-|\phi|^2}|X|^{2n-2}D^4 \exp\operatorname{tr}\log\big(\mathbb{I} - |X|^{-2}N\big)
\end{aligned}
$$

- (1) : The variables X and ϕ_I are dummy variables corresponding to a continuous limit of the flux quanta, where Z from Denef-Douglas is written as ϕ to prevent confusion with the complex moduli z. Each are complex variables, integrated over the entire complex plane.
  - In the orthogonal frame, raising and lowering indices is trivial. The convention here is that holomorphic variables in lower indices, and antiholo in upper. $|\phi|^2 = \sum_I \phi_I\overline{\phi}^I$.
- (2) : Evaluation of the determinant of the block matrix, from

$$\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det\big(D - BA^{-1}C\big).$$

- (3) : The determinant of a rank one perturbation to identity is given as

$$\det\Big(\mathbb{I} - \overline{\phi}^I\phi_J/|X|^2\Big) = 1 - \phi_I\overline{\phi}^I/|X|^2 = 1 - |\phi|^2/|X|^2.$$

  In addition, the inverse of a rank one perturbation to identity is given by the Sherman-Morrison formula

$$(\mathbb{I} - \phi_J\overline{\phi}^K/|X|^2)^{-1} = \mathbb{I} + (\overline{\phi}^J\phi_K)/(|X|^2 - |\phi|^2).$$

- (4) : $D^2 \coloneqq |X|^2 - |\phi|^2$.
- (5) : $\log\det M = \operatorname{tr}\log M$. All non-identity terms are collected into the matrix N.

$$
\begin{aligned}
N_I^P = & \ \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JPN}\overline{\phi}^K\phi_N + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}\overline{\phi}^K\overline{\phi}^P\phi_M\phi_N/D^2 \\
& + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LPN}\overline{\phi}^J\overline{\phi}^K\phi_L\phi_N/D^2 + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}\overline{\phi}^J\overline{\phi}^K\overline{\phi}^P\phi_L\phi_M\phi_N/D^4
\end{aligned}
$$

We note that in (4), each component of the matrix N_I^P inside the determinant is a degree 1 polynomial with respect to terms of the form $\mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}$. Therefore, the determinant itself will be a degree n polynomial wrt. $\mathcal{F}\bar{\mathcal{F}}$, and the following proposition holds;

Proposition 1. The integrand can be exactly obtained by expanding $\exp\operatorname{tr}\log\big(\mathbb{I} - |X|^{-2}N\big)$ up to n-th degree.

$$
\begin{aligned}
N_I^P &= \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JPN}\overline{\phi}^K\phi_N + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}\overline{\phi}^K\overline{\phi}^P\phi_M\phi_N/D^2 \\
&\quad + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LPN}\overline{\phi}^J\overline{\phi}^K\phi_L\phi_N/D^2 + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}\overline{\phi}^J\overline{\phi}^K\overline{\phi}^P\phi_L\phi_M\phi_N/D^4
\end{aligned}
$$

The objective is evaluating ρ up to the d-th degree in N for $d \leq n$. This article is divided into three sections; the first evaluates ρ for degrees $d = 1$ and $d = 2$ to demonstrate the general process, the second demonstrates a diagrammatic/combinatoric method, and the last provides proofs for the propositions in the article.

$d = 1$

Step 1 First, the expression $\exp\operatorname{tr}\log\big(\mathbb{I} - |X|^{-2}N\big)$ is evaluated up to 1st order in N.

$$
\log\big(1 - |X|^{-2}N\big) = -|X|^{-2}N - \frac{1}{2}|X|^{-4}N^2 - \cdots
$$

$$
\begin{aligned}
\exp\operatorname{tr}\log\big(1 - |X|^{-2}N\big) &= \exp\left(-|X|^{-2}\operatorname{tr}N - \frac{1}{2}|X|^{-4}\operatorname{tr}N^2 - \cdots\right) \\
&= 1 - |X|^{-2}\operatorname{tr}N + \cdots = 1 - |X|^{-2}N_I^I + \cdots
\end{aligned}
$$

The first order contribution to the integrand is

$$
\begin{aligned}
&e^{-|X|^2-|\phi|^2}|X|^{2n-2}\,D^4\cdot|X|^{-2}N_I^I \\
=&e^{-|X|^2-|\phi|^2}|X|^{2n-2}\,D^4\cdot|X|^{-2}\Big(\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JIN}\overline{\phi}^K\phi_N + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}\overline{\phi}^K\overline{\phi}^I\phi_M\phi_N/D^2 \\
&\qquad + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LIN}\overline{\phi}^J\overline{\phi}^K\phi_L\phi_N/D^2 + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}\overline{\phi}^J\overline{\phi}^K\overline{\phi}^I\phi_L\phi_M\phi_N/D^4\Big) \\
=&e^{-|X|^2-|\phi|^2}|X|^{2n-4}\Big(\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JIN}\overline{\phi}^K\phi_N D^4 + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}\overline{\phi}^K\overline{\phi}^I\phi_M\phi_N D^2 \\
&\qquad + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LIN}\overline{\phi}^J\overline{\phi}^K\phi_L\phi_N D^2 + \mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}\overline{\phi}^J\overline{\phi}^K\overline{\phi}^I\phi_L\phi_M\phi_N\Big)
\end{aligned}
$$

Step 2 Next, integrate out all $\phi_I, \overline{\phi}^J$. The Gaussian integral $\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\overline{\phi}^J$ is only nonzero when $I = J$, and equals δ_I^J. When there are multiple pairs of $\phi, \overline{\phi}$, the final result is given by Wick's theorem, and corresponds to all possible ways to contract over the given indices;

$$
\begin{aligned}
&\pi^{-n}\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\overline{\phi}^J = \delta_I^J \\
&\pi^{-n}\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\phi_J\overline{\phi}^K\overline{\phi}^L = \delta_I^K\delta_J^L + \delta_I^L\delta_J^K \\
&\pi^{-n}\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\phi_J\phi_K\overline{\phi}^L\overline{\phi}^M\overline{\phi}^N = \delta_I^L\delta_J^M\delta_K^N + \delta_I^L\delta_J^N\delta_K^M + \delta_I^M\delta_J^N\delta_K^L + \delta_I^M\delta_J^L\delta_K^N + \delta_I^N\delta_J^L\delta_K^M + \delta_I^N\delta_J^M\delta_K^L
\end{aligned}
$$

There are also D^2 and D^4 terms involved, which are functions of $\phi, \overline{\phi}$; these terms are a scalar multiple of the contraction terms for the D^0 case above;

$$
\delta_K^K = \delta_K^L\delta_L^K = n,\ \delta_I^K\delta_K^J = \delta_I^K\delta_K^L\delta_L^J = \delta_I^J
$$

$$
\begin{aligned}
\pi^{-n}\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\overline{\phi}^J D^4 &= \int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\overline{\phi}^J(|X|^4 - 2|X|^2\phi_K\overline{\phi}^K + \phi_K\phi_L\overline{\phi}^K\overline{\phi}^L) \\
&= |X|^4\delta_I^J - 2|X|^2(\delta_I^J\delta_K^K + \delta_I^K\delta_K^J) + (\delta_I^J\delta_K^K\delta_L^L + \delta_I^J\delta_K^L\delta_L^K + \delta_I^K\delta_K^L\delta_L^J + \delta_I^K\delta_K^J\delta_L^L + \delta_I^L\delta_K^J\delta_L^K + \delta_I^L\delta_K^K\delta_L^J) \\
&= |X|^4\delta_I^J - 2|X|^2(\delta_I^J\cdot n + \delta_I^J) + (\delta_I^J\cdot n^2 + \delta_I^J\cdot n + \delta_I^J + \delta_I^J\cdot n + \delta_I^J + \delta_I^J\cdot n) \\
&= \delta_I^J(|X|^4 - 2(n+1)|X|^2 + n^2 + 3n + 2)
\end{aligned}
$$

$$
\begin{aligned}
\pi^{-n}\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\phi_J\overline{\phi}^K\overline{\phi}^L D^2 &= \int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\phi_I\phi_J\overline{\phi}^K\overline{\phi}^L(|X|^2 - \phi_M\overline{\phi}^M) \\
&= |X|^2(\delta_I^K\delta_J^L + \delta_I^L\delta_J^K) - (\delta_I^K\delta_J^L\delta_M^M + \delta_I^K\delta_J^M\delta_M^L + \delta_I^L\delta_J^M\delta_M^K + \delta_I^L\delta_J^K\delta_M^M + \delta_I^M\delta_J^K\delta_M^L + \delta_I^M\delta_J^L\delta_M^K) \\
&= |X|^2(\delta_I^K\delta_J^L + \delta_I^L\delta_J^K) - (\delta_I^K\delta_J^L\cdot n + \delta_I^K\delta_J^L + \delta_I^L\delta_J^K + \delta_I^L\delta_J^K\cdot n + \delta_I^L\delta_J^K + \delta_I^K\delta_J^L) \\
&= (\delta_I^K\delta_J^L + \delta_I^L\delta_J^K)(|X|^2 - (n+2))
\end{aligned}
$$

Then, evaluating the total integral,

$$
\begin{aligned}
&\pi^{-2(n+1)}e^{-|X|^2}|X|^{2n-4}\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\Big(\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JIN}\overline{\phi}^K\phi_N D^4+\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}\overline{\phi}^K\overline{\phi}^I\phi_M\phi_N D^2 \\
&\qquad\qquad +\mathcal{F}_{IJK}\bar{\mathcal{F}}^{LIN}\overline{\phi}^J\overline{\phi}^K\phi_L\phi_N D^2+\mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}\overline{\phi}^J\overline{\phi}^K\overline{\phi}^I\phi_L\phi_M\phi_N\Big) \\
=&\pi^{-(n+2)}e^{-|X|^2}|X|^{2n-4}[\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JIN}\delta^K_N(|X|^4-2(n+1)|X|^2+n^2+3n+2) \\
&\qquad +\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}(\delta^K_M\delta^I_N+\delta^I_M\delta^K_N)(|X|^2-(n+2)) \\
&\qquad +\mathcal{F}_{IJK}\bar{\mathcal{F}}^{LIN}(\delta^J_L\delta^K_N+\delta^K_L\delta^J_N)(|X|^2-(n+2)) \\
&\qquad +\mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}(\delta^L_I\delta^M_J\delta^N_K+\delta^L_I\delta^N_J\delta^M_K+\delta^M_I\delta^N_J\delta^L_K+\delta^M_I\delta^L_J\delta^N_K+\delta^N_I\delta^L_J\delta^M_K+\delta^N_I\delta^M_J\delta^L_K)]
\end{aligned}
$$

Step 3 Third, contract indices on $\mathcal{F}\bar{\mathcal{F}}$ into scalar $U(n)$ invariants. For $d=1$, there is only one invariant $\mathcal{F}_{IJK}\bar{\mathcal{F}}^{IJK}=|\mathcal{F}|^2$. Then, integrate out $|X|$.

$$
\begin{aligned}
&\pi^{-(n+2)}\int \mathrm{d}^2X e^{-|X|^2}|X|^{2n-4}|\mathcal{F}|^2[(|X|^4-2(n+1)|X|^2+n^2+3n+2)+4(|X|^2-(n+2))+6] \\
=&\pi^{-(n+2)}|\mathcal{F}|^2\int \mathrm{d}^2X e^{-|X|^2}|X|^{2n-4}[|X|^4+(-2n+2)|X|^2+n^2-n] \\
=&\pi^{-(n+2)}|\mathcal{F}|^2(\langle|X|^{2n}\rangle+(-2n+2)\langle|X|^{2n-2}\rangle+(n^2-n)\langle|X|^{2n-4}\rangle)
\end{aligned}
$$

Here, it is noted that for $n=1$, the final term $\langle|X|^{2n-4}\rangle=\pi(-1)!$ is a singular term, but its total coefficient is 0. This is reflected in the fact that for $n=1$, there is no singular term in the integrand for the initial Douglas-Denef equation. For higher orders, the proposition below (which is proved in a later section) is used;

Proposition 2. Expanding the integrand of ρ in powers of $|X|$, the only singular term is proportional to $|X|^{-2}$, and its coefficient is 0.

$d=2$

Step 1 First, the expression $\exp\operatorname{tr}\log\big(\mathbb{I}-|X|^{-2}N\big)$ is evaluated up to 2nd order in N.

$$
\log\big(\mathbb{I}-|X|^{-2}N\big)=-|X|^{-2}N-\frac{1}{2}|X|^{-4}N^2-\cdots
$$

$$
\begin{aligned}
\exp\operatorname{tr}\log\big(\mathbb{I}-|X|^{-2}N\big)&=\exp\left(-|X|^{-2}\operatorname{tr}N-\frac{1}{2}|X|^{-4}\operatorname{tr}N^2-\cdots\right) \\
&=\cdots-\frac{1}{2}|X|^{-4}\operatorname{tr}N^2+\frac{1}{2}|X|^{-4}(\operatorname{tr}N)^2\cdots \\
&=\cdots-\frac{1}{2}|X|^{-4}N^J_IN^I_J+\frac{1}{2}|X|^{-4}N^I_IN^J_J\cdots
\end{aligned}
$$

Both second order contributions are of the form $N^{P_1}_{I_1}N^{P_2}_{I_2}$ with "external indices" contracted via

$$
\begin{aligned}
N^J_IN^I_J&=N^{P_1}_{I_1}N^{P_2}_{I_2}\delta^{P_2}_{I_1}\delta^{P_1}_{I_2}, \\
N^I_IN^J_J&=N^{P_1}_{I_1}N^{P_2}_{I_2}\delta^{P_1}_{I_1}\delta^{P_2}_{I_2}.
\end{aligned}
$$

The two contractions above are all the possible contractions of $N^{P_1}_{I_1}N^{P_2}_{I_2}$, and the final contribution is a sum over them with a total prefactor of $\frac{\pm1}{2!}$. While it isn't clear yet, the proposition below (which is proved in a later section) is used for higher orders;

Proposition 3. In the series expansion of $\exp\operatorname{tr}\log\big(\mathbb{I}-|X|^{-2}N\big)$, the d-th order term equals the sum over all possible contractions of external indices, with a prefactor $(-1)^{\mathrm{disj}(\sigma)}/d!$, where $\mathrm{disj}(\sigma)$ refers to the number of disjoint cycles in the permutation $\sigma: I_i\to$ (index I_i is contracted with).

For a general prefactor A, the contribution of $N_{I_1}^{P_1} N_{I_2}^{P_2}$ is

$$
\begin{aligned}
&e^{-|X|^2-|\phi|^2}|X|^{2n-2}D^4 \cdot |X|^{-4}\left(AN_{I_1}^{P_1}N_{I_2}^{P_2}\right)\\
=&e^{-|X|^2-|\phi|^2}|X|^{2n-6}A\Big(\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1P_1N_1}\mathcal{F}_{I_2J_2K_2}\bar{\mathcal{F}}^{J_2P_2N_2}\overline{\phi}^{K_1}\phi_{N_1}\overline{\phi}^{K_2}\phi_{N_2}D^4\\
&+2\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1P_1N_1}\mathcal{F}_{I_2J_2K_2}\bar{\mathcal{F}}^{J_2M_2N_2}\overline{\phi}^{K_1}\phi_{N_1}\overline{\phi}^{K_2}\overline{\phi}^{I}\phi_{M_2}\phi_{N_2}D^2+2\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1P_1N_1}\mathcal{F}_{I_2J_2K_2}\bar{\mathcal{F}}^{L_2P_2N_2}\overline{\phi}^{K_1}\phi_{N_1}\overline{\phi}^{J_2}\overline{\phi}^{K_2}\phi_{L_2}\phi_{N_2}D^2\\
&+\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1M_1N_1}\mathcal{F}_{I_2J_2K_2}\bar{\mathcal{F}}^{J_2M_2N_2}\overline{\phi}^{K_1}\overline{\phi}^{P_1}\phi_{M_1}\phi_{N_1}\overline{\phi}^{K_2}\overline{\phi}^{P_2}\phi_{M_2}\phi_{N_2}\\
&+2\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1M_1N_1}\mathcal{F}_{I_2J_2K_2}\bar{\mathcal{F}}^{L_2P_2N_2}\overline{\phi}^{K_1}\overline{\phi}^{P_1}\phi_{M_1}\phi_{N_1}\overline{\phi}^{J_2}\overline{\phi}^{K_2}\phi_{L_2}\phi_{N_2}+\cdots\Big)
\end{aligned}
$$

Each term in the expanded expression precisely corresponds to a combination with repetition chosen from the 4 terms in N_I^P. Each term also has two lower indices I_1, I_2 and two upper indices P_1, P_2, which summing over all contractions (with appropriate prefactors) will give the total contribution to the integrand. Finally, each term has a number of "internal indices" attached to the variables $\phi, \overline{\phi}$, which evaluating the integral in Step 2 is equivalent to summing over all contractions.

Step 2 The same method as in $d = 1$ is used to integrate out the variables $\phi, \overline{\phi}$ via contractions. However, unlike $d = 1$, there are terms that are singular in D, corresponding to the terms in $\exp \operatorname{tr} \log\big(\mathbb{I} - |X|^{-2}N\big)$ proportional to D^{-6} or worse. Attempting to regulate these singularities is an ill-defined problem. Instead, the following proposition (which is proved in a later section) is used;

Proposition 4. Expanding the integrand of ρ as a formal series in $D^2 = |X|^2 - |\phi|^2$, the singular coefficients vanish.

$$N_I^P = N_{(I)}{}_I^P + N_{(II)}{}_I^P D^{-2} + N_{(III)}{}_I^P D^{-2} + N_{(IV)}{}_I^P D^{-4}$$

Writing as above, there are only 7 remaining terms

$$\underbrace{N_{(I)}N_{(IV)}, N_{(II)}N_{(II)}, N_{(II)}N_{(III)}, N_{(III)}N_{(III)}}_{D^{-4}}, \underbrace{N_{(I)}N_{(II)}, N_{(I)}N_{(III)}}_{D^{-2}}, \underbrace{N_{(I)}N_{(I)}}_{D^0}$$

and this is true for any $d \geq 2$ (higher order terms can only include more $N_{(I)}$).
For demonstration purposes, the first term in the expanded equation above for the contraction $N_I^J N_J^I$ is computed;

$$
\begin{aligned}
&\pi^{-2(n+1)}e^{-|X|^2}|X|^{2n-6}\cdot -\frac{1}{2}\int \mathrm{d}^{2n}\phi e^{-|\phi|^2}\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1P_1N_1}\mathcal{F}_{P_1J_2K_2}\bar{\mathcal{F}}^{J_2I_1N_2}\overline{\phi}^{K_1}\phi_{N_1}\overline{\phi}^{K_2}\phi_{N_2}D^4\\
=&\pi^{-(n+2)}e^{-|X|^2}|X|^{2n-6}\cdot -\frac{1}{2}\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1P_1N_1}\mathcal{F}_{P_1J_2K_2}\bar{\mathcal{F}}^{J_2I_1N_2}(|X|^4+(-2n-4)|X|^2+(n^2+5n+8))(\delta_{N_1}^{K_1}\delta_{N_2}^{K_2}+\delta_{N_2}^{K_1}\delta_{N_1}^{K_2}),
\end{aligned}
$$

Step 3 For demonstration purposes, contract indices on $\mathcal{F}\bar{\mathcal{F}}$ for the term above. For $d = 2$, there are two possible invariants, $|\mathcal{F}|^4 = \mathcal{F}_{(3,0)} = \mathcal{F}_{IJK}\mathcal{F}_{LMN}\bar{\mathcal{F}}^{IJK}\bar{\mathcal{F}}^{LMN}$ and $\mathcal{F}_{(2,1)} = \mathcal{F}_{IJK}\mathcal{F}_{LMN}\bar{\mathcal{F}}^{LJK}\bar{\mathcal{F}}^{IMN}$.

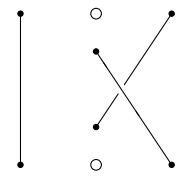
$$
\begin{aligned}
&\pi^{-(n+2)}\int \mathrm{d}^2X e^{-|X|^2}|X|^{2n-6}\cdot -\frac{1}{2}\mathcal{F}_{I_1J_1K_1}\bar{\mathcal{F}}^{J_1P_1N_1}\mathcal{F}_{P_1J_2K_2}\bar{\mathcal{F}}^{J_2I_1N_2}(|X|^4+(-2n-4)|X|^2+(n^2+5n+8))(\delta_{N_1}^{K_1}\delta_{N_2}^{K_2}+\delta_{N_2}^{K_1}\delta_{N_1}^{K_2})\\
&=\pi^{-(n+2)}\cdot -\frac{1}{2}(\mathcal{F}_{(2,1)}+\mathcal{F}_{(2,1)})\int \mathrm{d}^2X e^{-|X|^2}(|X|^{2n-2}+(-2n-4)|X|^{2n-4}+(n^2+5n+8)|X|^{2n-6})\\
&=-\pi^{-(n+2)}\mathcal{F}_{(2,1)}(\langle|X|^{2n-2}\rangle+(-2n-4)\langle|X|^{2n-4}\rangle+(n^2+5n+8)\langle|X|^{2n-6}\rangle)
\end{aligned}
$$

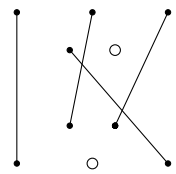
Due to Proposition 1, there is no need to consider $n = 1$ for $d = 2$. Due to Proposition 2, the possible singular term at $n = 2$, $\langle|X|^{2n-6}\rangle$ has vanishing coefficient when adding together all coefficients.

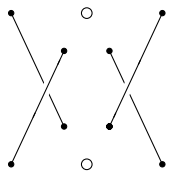
Diagrammatic Method for Order d

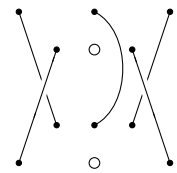
The combinatoric nature of the process above can be utilised to introduce a diagrammatic representation. For each one of the 4 terms in N_I^P, the lower indices of $\mathcal{F}$ are represented as "outer" lower dots, upper indices of $\bar{\mathcal{F}}$ as "outer" upper dots, lower indices of ϕ as "inner" lower dots, and upper indices of $\overline{\phi}$ as "inner" upper dots.

Contraction of two indices is represented as a line connecting the two corresponding dots. Dots corresponding to "external indices" (which will have no lines connected to them in the "noncontracted" state) are denoted as a circle for emphasis. Importantly, as $\mathcal{F}_{IJK}$ is totally symmetric, the "outer" upper dots can be permuted in any order;

$\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JPN}\overline{\phi}^{K}\phi_N =$ 

$\mathcal{F}_{IJK}\bar{\mathcal{F}}^{JMN}\overline{\phi}^{K}\overline{\phi}^{P}\phi_M\phi_N D^{-2} =$  D^{-2}

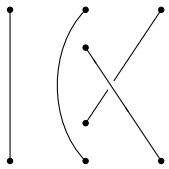
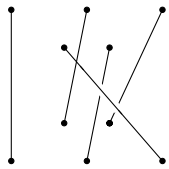
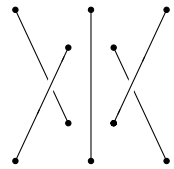
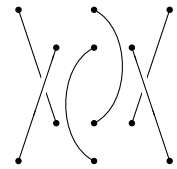
$\mathcal{F}_{IJK}\bar{\mathcal{F}}^{LPN}\overline{\phi}^{J}\overline{\phi}^{K}\phi_L\phi_N D^{-2} =$  D^{-2}

$\mathcal{F}_{IJK}\bar{\mathcal{F}}^{LMN}\overline{\phi}^{J}\overline{\phi}^{K}\overline{\phi}^{P}\phi_L\phi_M\phi_N D^{-4} =$  D^{-4}

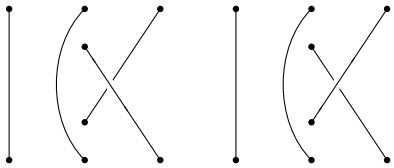
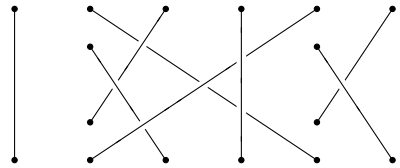
Contracting External Indices

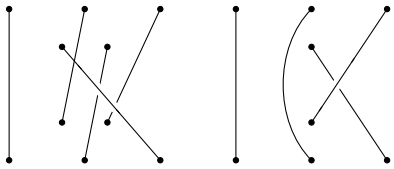
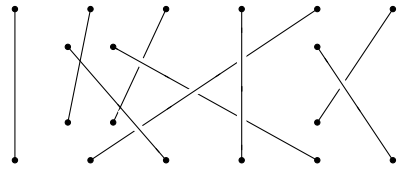
Calculating the order d contribution to the integrand can be done by drawing all combinations of d diagrams from the 4 in N, and drawing all possible contractions of lower and upper circles. Only diagrams with singular degree up to D^{-4} need to be considered.

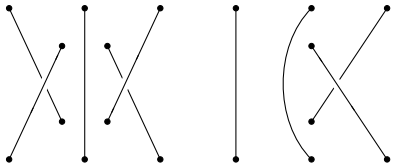
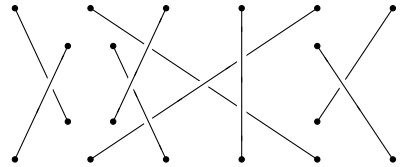
$d = 1$

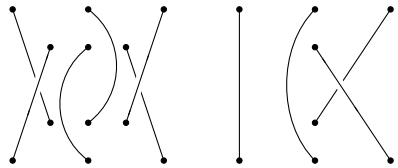
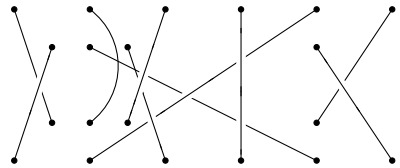
$-|X|^{2n-4}D^4$  , $-|X|^{2n-4}D^2$  , $-|X|^{2n-4}D^2$  , $-|X|^{2n-4}$ 

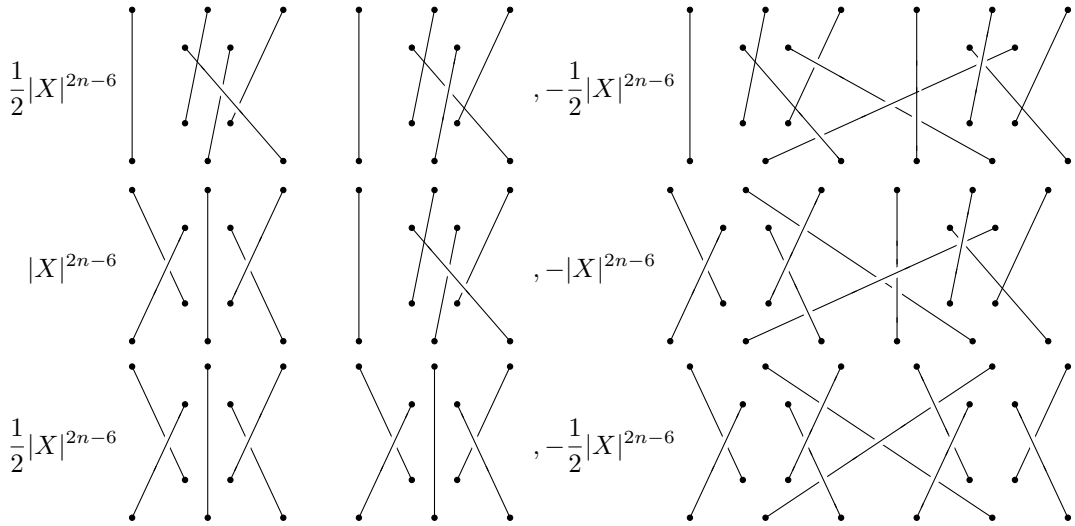
$d = 2$

$\frac{1}{2}|X|^{2n-6}D^4$  , $-\frac{1}{2}|X|^{2n-6}D^4$ 

$|X|^{2n-6}D^2$  , $-|X|^{2n-6}D^2$ 

$|X|^{2n-6}D^2$  , $-|X|^{2n-6}D^2$ 

$|X|^{2n-6}$  , $-|X|^{2n-6}$ 



Contracting Inner Indices

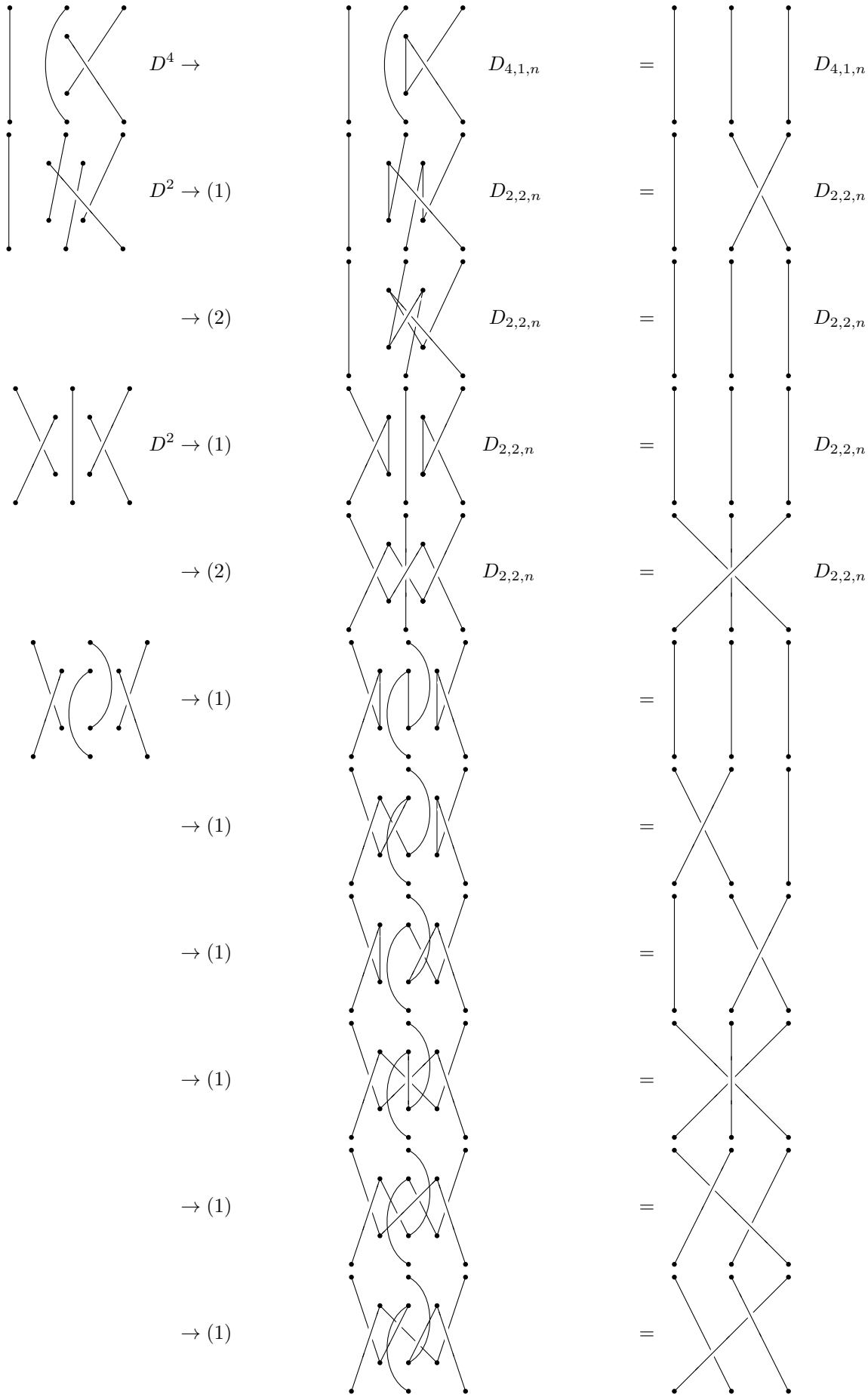
The integrand is integrated over all variables ϕ by performing Wick contraction; for each diagram from the previous step, draw all possible contractions of lower internal dots with upper internal dots, with all combinatoric coefficients carrying over. Then erase all internal dots to leave only connections between external dots. If there is D^2 or D^4 in the term, additional contact terms must be considered;

$$\begin{aligned}
\left\langle \overline{\phi}^{A_1} \cdots \overline{\phi}^{A_m} \phi_{B_1} \cdots \phi_{B_m} D^2 \right\rangle &= (\text{all contractions}) \left\langle D^2 \right\rangle + \sum (\text{all contractions not involving } A_i, B_j) \left\langle \partial_{A_i} \partial^{B_j} D^2 \right\rangle \\
&= (\text{all contractions}) \cdot (-n + |X|^2) - \sum (\text{all contractions not involving } A_i, B_j) \cdot \delta_{A_i}^{B_j} \\
&= (\text{all contractions}) \cdot \left(-n - \frac{m^2 \cdot (m-1)!}{m!} + |X|^2 \right) \\
&= (\text{all contractions}) \cdot (-n - m + |X|^2) \coloneqq (\text{all contractions}) \cdot D_{2,m,n}
\end{aligned}$$

$$\begin{aligned}
\left\langle \overline{\phi}^{A_1} \cdots \overline{\phi}^{A_m} \phi_{B_1} \cdots \phi_{B_m} D^4 \right\rangle &= (\text{all contractions}) \left\langle D^4 \right\rangle \\
&+ \sum (\text{all contractions not involving } A_i, B_j) \left\langle \partial_{A_i} \partial^{B_j} D^4 \right\rangle \\
&+ \sum (\text{all contractions not involving } A_i, A_k, B_j, B_l) \left\langle \partial_{A_i} \partial_{A_k} \partial^{B_j} \partial^{B_l} D^4 \right\rangle \\
&= (\text{all contractions}) \cdot \left[2 \cdot n + n(n-1) - 2n|X|^2 + |X|^4 \right] \\
&+ \sum (\text{all contractions not involving } A_i, B_j) \cdot \delta_{B_j}^{A_i} \cdot (2n + 2 - 2|X|^2) \\
&+ \sum (\text{all contractions not involving } A_i, A_k, B_j, B_l) \cdot (\delta_{B_j}^{A_i} \delta_{B_l}^{A_k} + \delta_{B_l}^{A_i} \delta_{B_j}^{A_k}) \\
&= (\text{all contractions}) \cdot [(n^2 + n - 2n|X|^2 + |X|^4) + (2n + 2 - 2|X|^2)m + m(m-1)] \\
&= (\text{all contractions}) \cdot [(n^2 + n + m(2n+2) + m(m-1)) + (-2n - 2m)|X|^2 + |X|^4] \\
&\coloneqq (\text{all contractions}) \cdot D_{4,m,n}
\end{aligned}$$

The contact terms are only multiplicative in nature, and be exchanged with D^2 or D^4 respectively. Then, collect all terms wrt. powers in $|X|^2$, then integrate over X with $\left\langle |X|^{2n} \right\rangle = n!$.

$\boldsymbol{d = 1}$



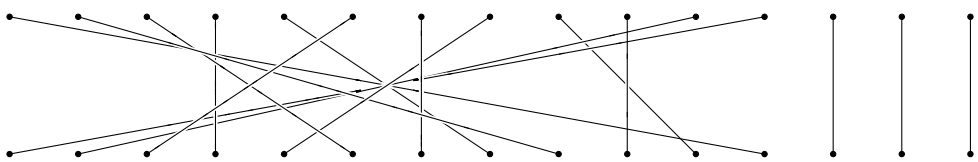
Contracting Outer Indices : Classification of Diagrams into Invariants of $U(n)$

The resulting diagrams are bipartite graphs, and represent invariants of $U(n)$ that arise from contracting d copies of $\mathcal{F}$ and $\bar{\mathcal{F}}$. A large number of permutations create equivalent invariants; first note the term of the form

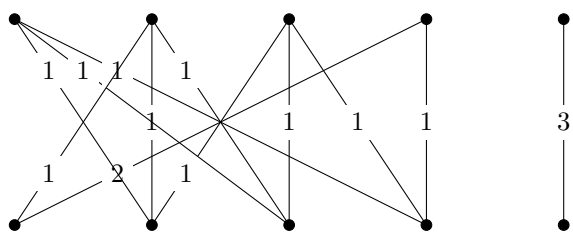
$$\mathcal{F}_{I_1 J_1 K_1} \cdots \mathcal{F}_{I_d J_d K_d} \bar{\mathcal{F}}^{I'_1 J'_1 K'_1} \ldots \bar{\mathcal{F}}^{I'_d J'_d K'_d}$$

is identical under permutations between any (I_i, J_i, K_i) and (I_j, J_j, K_j), between any (I'_i, J'_i, K'_i) and (I'_j, J'_j, K'_j), or within any (I_i, J_i, K_i) or (I'_i, J'_i, K'_i) (due to total symmtery of $\mathcal{F}$).

The permutations within each block (within (I_i, J_i, K_i) and vice versa) can be factored out by representing the bipartite graph as a weighted bipartite graph of d upper and lower vertices each, where the edge connecting the lower vertex i and upper vertex j are the number of elements mapped from block i to block j in the initial graph.



For example, the diagram above can be represents a $d = 5$ diagram, which would be represented by the graph



Then, the permutations between $\mathcal{F}$ or $\bar{\mathcal{F}}$ blocks correspond to graph isomorphisms. The problem of converting the initial bipartite graph into the weighted bipartite graph, and classifying each weighted bipartite graph by equivalence class with respect to graph isomorphisms is implemented using `igraph`, and is at least a $\mathcal{O}(d!)$ process taking up the majority of runtime.

It is still possible that different isomoprhism classes of the weighted bipartite graphs are equivalent scalar invariants, but a further reduction is not explored in this article. Numerically, it can be shown that the two invariants for $d = 2$ and five invariants for $d = 3$ differ given a sufficiently general setup (for example, $\mathcal{F}_{IJK}$ given by $[1, 1, 1, 6, 9]$).

$Finally, notation to denote these graphs are introduced, as a matrix where each column denotes the connection weights for each lower node. Then the diagram above would be in the form

$$((0, 1, 0, 2, 0), (1, 1, 1, 0, 0), (1, 1, 1, 0, 0), (1, 0, 1, 1, 0), (0, 0, 0, 0, 3)).$$

Proofs of Propositions

Proof of Proposition 4. Assume that when formally expanding the integrand as a series in D, there are singular terms with nonzero coefficients. Then for any fixed value of $|X|^2$ and any path $|\phi|^2 \to |X|^2$, the limit of the integrand along this path would be divergent. Fix such a path, $\phi = \overline{\phi} = a(|X|, 0, \cdots)$ such that the limit is realised by $a \to 1$. Then,

$$\begin{pmatrix} \delta^I_J \overline{X} - \overline{\phi}^I \phi_J / X & \mathcal{F}_{IJK} \overline{\phi}^K \\ \bar{\mathcal{F}}^{IJK} \phi_K & \delta^J_I X - \phi_I \overline{\phi}^J / \overline{X} \end{pmatrix} = \begin{pmatrix} \overline{X} - a^2|X|^2/X & 0 & \cdots & 0 & & & & \\ 0 & \overline{X} & \cdots & 0 & & \mathcal{F}_{IJ1} a|X| & & \\ \vdots & \vdots & \ddots & 0 & & & & \\ 0 & 0 & 0 & \overline{X} & & & & \\ & & & & X - a^2|X|^2/\overline{X} & 0 & \cdots & 0 \\ & \bar{\mathcal{F}}^{IJ1} a|X| & & & 0 & X & \cdots & 0 \\ & & & & \vdots & \vdots & \ddots & 0 \\ & & & & 0 & 0 & 0 & X \end{pmatrix}$$

$$\to \begin{pmatrix} 0 & 0 & \cdots & 0 & & & & \\ 0 & \overline{X} & \cdots & 0 & & \mathcal{F}_{IJ1}|X| & & \\ \vdots & \vdots & \ddots & 0 & & & & \\ 0 & 0 & 0 & \overline{X} & & & & \\ & & & & 0 & 0 & \cdots & 0 \\ & \bar{\mathcal{F}}^{IJ1}|X| & & & 0 & X & \cdots & 0 \\ & & & & \vdots & \vdots & \ddots & 0 \\ & & & & 0 & 0 & 0 & X \end{pmatrix}$$

However, the determinant of the matrix above cannot be divergent for any value of $|X|$, as it is a finite degree polynomial. Therefore, there cannot be any singular terms in the formal series of the integrand.

Proof of Proposition 2.

$$\det \begin{pmatrix} \delta^I_J \overline{X} - \overline{\phi}^I \phi_J / X & \mathcal{F}_{IJK} \overline{\phi}^K \\ \bar{\mathcal{F}}^{IJK} \phi_K & \delta^J_I X - \phi_I \overline{\phi}^J / \overline{X} \end{pmatrix}$$

The matrix above is manifestly invariant under $U(n)$ rotations

$$\phi_I \to U^J_I \phi_J,\ \overline{\phi}^I \to U^{\dagger I}{}_J \overline{\phi}^J,\ \mathcal{F}_{IJK} \to U^{I'}_I U^{J'}_J U^{K'}_K \mathcal{F}_{I'J'K'},$$

and so is the exponent $e^{-|\phi|^2}$. Consider the rotation such that $\phi_I \to (|\phi|, 0, \cdots), \overline{\phi}_I \to (|\phi|, 0, \cdots)$ and define $\mathcal{F}_{IJK} \to \mathcal{F}'_{IJK}$.

$$\begin{pmatrix} \delta^I_J \overline{X} - \overline{\phi}^I \phi_J / X & \mathcal{F}_{IJK} \overline{\phi}^K \\ \bar{\mathcal{F}}^{IJK} \phi_K & \delta^J_I X - \phi_I \overline{\phi}^J / \overline{X} \end{pmatrix} = \begin{pmatrix} \overline{X} - |\phi|^2/X & 0 & \cdots & 0 & & & & \\ 0 & \overline{X} & \cdots & 0 & & \mathcal{F}'_{IJ1}|\phi| & & \\ \vdots & \vdots & \ddots & 0 & & & & \\ 0 & 0 & 0 & \overline{X} & & & & \\ & & & & X - |\phi|^2/\overline{X} & 0 & \cdots & 0 \\ & \bar{\mathcal{F}}'^{IJ1}|\phi| & & & 0 & X & \cdots & 0 \\ & & & & \vdots & \vdots & \ddots & 0 \\ & & & & 0 & 0 & 0 & X \end{pmatrix}$$

The determinant of the matrix above then can at most have a singularity of $1/X\overline{X} = |X|^{-2}$, which is cancelled out by the $|X|^2$ term in the integrand. Therefore, there is no singular term in $|X|$ in the integrand.

Proof of Proposition 3. To motivate the general construction, the example of $d = 3$ is considered;

$$\begin{aligned} \exp \operatorname{tr} \log\big(\mathbb{I} - |X|^{-2} N\big) &= \exp\left(-|X|^{-2} \operatorname{tr} N - \frac{1}{2}|X|^{-4} \operatorname{tr} N^2 - \frac{1}{3}|X|^{-6} \operatorname{tr} N^3 - \cdots\right) \\ &= \cdots - \frac{1}{3}|X|^{-6} \operatorname{tr} N^3 + \frac{1}{2!} \cdot \frac{1}{2} \cdot 2|X|^{-6} \operatorname{tr} N \operatorname{tr} N^2 - \frac{1}{3!}|X|^{-6} (\operatorname{tr} N)^3 + \cdots \\ &= \cdots - \frac{1}{3}|X|^{-6} \operatorname{tr} N^3 + \frac{1}{2}|X|^{-6} \operatorname{tr} N \operatorname{tr} N^2 - \frac{1}{6}|X|^{-6} (\operatorname{tr} N)^3 + \cdots \end{aligned}$$

$$
\begin{aligned}
N^{P_1}_{I_1} N^{P_2}_{I_2} N^{P_3}_{I_3} \delta^{P_1}_{I_1} \delta^{P_2}_{I_2} \delta^{P_3}_{I_3} &= N^I_I N^J_J N^K_K = (\operatorname{tr} N)^3, && \operatorname{disj}(\sigma) = 3 \\
N^{P_1}_{I_1} N^{P_2}_{I_2} N^{P_3}_{I_3} \delta^{P_1}_{I_1} \delta^{P_3}_{I_2} \delta^{P_2}_{I_3} &= N^I_I N^K_J N^J_K = \operatorname{tr} N \operatorname{tr} N^2, && \operatorname{disj}(\sigma) = 2 \\
N^{P_1}_{I_1} N^{P_2}_{I_2} N^{P_3}_{I_3} \delta^{P_2}_{I_1} \delta^{P_3}_{I_2} \delta^{P_1}_{I_3} &= N^J_I N^K_J N^I_K = \operatorname{tr} N^3, && \operatorname{disj}(\sigma) = 1 \\
N^{P_1}_{I_1} N^{P_2}_{I_2} N^{P_3}_{I_3} \delta^{P_2}_{I_1} \delta^{P_1}_{I_2} \delta^{P_3}_{I_3} &= N^J_I N^I_J N^K_K = \operatorname{tr} N^2 \operatorname{tr} N, && \operatorname{disj}(\sigma) = 2 \\
N^{P_1}_{I_1} N^{P_2}_{I_2} N^{P_3}_{I_3} \delta^{P_3}_{I_1} \delta^{P_1}_{I_2} \delta^{P_2}_{I_3} &= N^K_I N^I_J N^J_K = \operatorname{tr} N^3, && \operatorname{disj}(\sigma) = 1 \\
N^{P_1}_{I_1} N^{P_2}_{I_2} N^{P_3}_{I_3} \delta^{P_3}_{I_1} \delta^{P_2}_{I_2} \delta^{P_1}_{I_3} &= N^K_I N^J_J N^I_K = \operatorname{tr} N \operatorname{tr} N^2, && \operatorname{disj}(\sigma) = 2
\end{aligned}
$$

The coefficient is accounted for combinatorically by the multiplicity of permutations generating the cycles corresponding to each trace term. Therefore, the final expression is a sum over all possible contractions with a prefactor $(-1)^{\operatorname{disj}(\sigma)}/d!$. The generalised proof is below;

$$
\begin{aligned}
&|X|^{2n-2} D^4 \exp \operatorname{tr} \log\big(\mathbb{I} - |X|^{-2} N\big) = |X|^{2n-2} D^4 \left(1 + \left(\sum_k -\frac{1}{k} |X|^{-2k} \operatorname{tr} N^k\right) + \frac{1}{2!}\left(\sum_k -\frac{1}{k} |X|^{-2k} \operatorname{tr} N^k\right)^2 + \cdots\right) \\
&=_{(1)} D^4 \sum_d \sum_{\substack{m=1,\cdots,d \\ k_1+\cdots+k_m=d}} \frac{1}{m!} \cdot \frac{|X|^{2n-2-2k_1\cdots-2k_m}(-1)^m}{k_1 k_2 \cdots k_m} \frac{m!}{\prod_k (\text{number of identical } k_i)!} \cdot \operatorname{tr} N^{k_1} \operatorname{tr} N^{k_2} \cdots \operatorname{tr} N^{k_m} \\
&=_{(2)} D^4 \sum_d \sum_{\substack{m=1,\cdots,d \\ k_1+\cdots+k_m=d}} \frac{|X|^{2n-2d-2}(-1)^m}{d!} N(k_1, \cdots, k_m) \cdot \operatorname{tr} N^{k_1} \operatorname{tr} N^{k_2} \cdots \operatorname{tr} N^{k_m} \\
&=_{(3)} D^4 \sum_d \sum_{\substack{m=1,\cdots,d \\ k_1+\cdots+k_m=d}} \frac{|X|^{2n-2d-2}(-1)^m}{d!} \cdot \sum_{\substack{\sigma \in S_d \\ \text{disj. cycles} \\ k_1,\cdots,k_d}} N^{I_{\sigma(1)}}_{I_1} \cdots N^{I_{\sigma(d)}}_{I_d} \\
&=_{(4)} D^4 \sum_d \sum_{\sigma \in S_d} \frac{|X|^{2n-2d-2}(-1)^{\operatorname{disj}(\sigma)}}{d!} N^{I_{\sigma(1)}}_{I_1} \cdots N^{I_{\sigma(d)}}_{I_d}
\end{aligned}
$$

- (1) : Obtained from directly expanding the order d coefficient of the m-th term in the $\exp \operatorname{tr} \log$ equation.
- (2) : $N(k_1, \cdots, k_m)$ is the number of permutations in S_d that have m disjoint cycles of length $k_1, \cdots, k_m$.

$$
N(k_1, \cdots, k_m) = \frac{d!}{k_1! k_2! \cdots k_m!} \frac{1}{\prod_k (\text{number of identical } k_i)!} \cdot (k_1 - 1)!(k_2 - 1)! \cdots (k_m - 1)!
$$

- (3) : If $\sigma \in S_d$ is a permutation with disjoint cycles of length $k_1, \cdots, k_m$, the contracted sum $N^{I_{\sigma(1)}}_{I_1} \cdots N^{I_{\sigma(d)}}_{I_d}$ equals $\operatorname{tr} N^{k_1} \cdots \operatorname{tr} N^{k_m}$.
- (4) : $\operatorname{disj}(\sigma)$ refers to the number of disjoint cycles in σ.

Results

Based on the notation introduced in the previous section, we classify all invariants up to order 5.

$$
\begin{aligned}
|\mathcal{F}|^4 &= \mathcal{F}_1^{(2)} = ((3,0),(0,3)), & \mathcal{F}_{(2,1)} &= \mathcal{F}_2^{(2)} = ((2,1),(1,2)) \\
|\mathcal{F}|^6 &= \mathcal{F}_1^{(3)} = ((3,0,0),(0,3,0),(0,0,3)), & |\mathcal{F}|^2\mathcal{F}_{(2,1)} &= \mathcal{F}_2^{(3)} = ((3,0,0),(0,2,1),(0,1,2)) \\
\mathcal{F}_3^{(3)} &= ((2,1,0),(0,2,1),(1,0,2)), & \mathcal{F}_4^{(3)} &= ((2,1,0),(1,1,1),(0,1,2)) \\
\mathcal{F}_5^{(3)} &= ((1,1,1),(1,1,1),(1,1,1))
\end{aligned}
$$

$$
\begin{aligned}
|\mathcal{F}|^8 = \mathcal{F}_1^{(4)} &= ((3,0,0,0),(0,3,0,0),(0,0,3,0),(0,0,0,3)) \\
|\mathcal{F}|^4\mathcal{F}_{(2,1)} = \mathcal{F}_2^{(4)} &= ((3,0,0,0),(0,3,0,0),(0,0,2,1),(0,0,1,2)) \\
|\mathcal{F}|^2\mathcal{F}_3^{(3)} = \mathcal{F}_3^{(4)} &= ((3,0,0,0),(0,2,1,0),(0,0,2,1),(0,1,0,2)) \\
\mathcal{F}_{(2,1)}^2 = \mathcal{F}_4^{(4)} &= ((2,1,0,0),(1,2,0,0),(0,0,2,1),(0,0,1,2)) \\
\mathcal{F}_5^{(4)} &= ((2,1,0,0),(0,2,1,0),(0,0,2,1),(1,0,0,2)) \\
|\mathcal{F}|^2\mathcal{F}_4^{(3)} = \mathcal{F}_6^{(4)} &= ((3,0,0,0),(0,2,1,0),(0,1,1,1),(0,0,1,2)) \\
\mathcal{F}_7^{(4)} &= ((2,1,0,0),(0,2,1,0),(1,0,1,1),(0,0,1,2)) \\
\mathcal{F}_8^{(4)} &= ((2,0,1,0),(0,2,0,1),(1,0,1,1),(0,1,1,1)) \\
|\mathcal{F}|^2\mathcal{F}_5^{(3)} = \mathcal{F}_9^{(4)} &= ((3,0,0,0),(0,1,1,1),(0,1,1,1),(0,1,1,1)) \\
\mathcal{F}_{10}^{(4)} &= ((2,1,0,0),(0,1,2,0),(1,0,1,1),(0,1,0,2)) \\
\mathcal{F}_{11}^{(4)} &= ((2,1,0,0),(0,1,1,1),(1,0,1,1),(0,1,1,1))
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_1^{(5)} &= ((3,0,0,0,0),(0,3,0,0,0),(0,0,3,0,0),(0,0,0,3,0),(0,0,0,0,3)) \\
\mathcal{F}_2^{(5)} &= ((3,0,0,0,0),(0,3,0,0,0),(0,0,3,0,0),(0,0,0,2,1),(0,0,0,1,2)) \\
\mathcal{F}_3^{(5)} &= ((3,0,0,0,0),(0,3,0,0,0),(0,0,2,1,0),(0,0,0,2,1),(0,0,1,0,2)) \\
\mathcal{F}_4^{(5)} &= ((3,0,0,0,0),(0,2,1,0,0),(0,1,2,0,0),(0,0,0,2,1),(0,0,0,1,2)) \\
\mathcal{F}_5^{(5)} &= ((3,0,0,0,0),(0,2,1,0,0),(0,0,2,1,0),(0,0,0,2,1),(0,1,0,0,2)) \\
\mathcal{F}_6^{(5)} &= ((2,1,0,0,0),(1,2,0,0,0),(0,0,2,1,0),(0,0,0,2,1),(0,0,1,0,2)) \\
\mathcal{F}_7^{(5)} &= ((2,1,0,0,0),(0,2,1,0,0),(0,0,2,1,0),(0,0,0,2,1),(1,0,0,0,2)) \\
\mathcal{F}_8^{(5)} &= ((3,0,0,0,0),(0,3,0,0,0),(0,0,2,1,0),(0,0,1,1,1),(0,0,0,1,2)) \\
\mathcal{F}_9^{(5)} &= ((3,0,0,0,0),(0,2,1,0,0),(0,0,2,1,0),(0,1,0,1,1),(0,0,0,1,2)) \\
\mathcal{F}_{10}^{(5)} &= ((3,0,0,0,0),(0,2,0,1,0),(0,0,2,0,1),(0,1,0,1,1),(0,0,1,1,1)) \\
\mathcal{F}_{11}^{(5)} &= ((2,1,0,0,0),(1,2,0,0,0),(0,0,2,1,0),(0,0,1,1,1),(0,0,0,1,2)) \\
\mathcal{F}_{12}^{(5)} &= ((2,1,0,0,0),(0,2,1,0,0),(0,0,2,1,0),(1,0,0,1,1),(0,0,0,1,2)) \\
\mathcal{F}_{13}^{(5)} &= ((2,1,0,0,0),(0,2,0,1,0),(0,0,2,0,1),(1,0,0,1,1),(0,0,1,1,1)) \\
\mathcal{F}_{14}^{(5)} &= ((3,0,0,0,0),(0,3,0,0,0),(0,0,1,1,1),(0,0,1,1,1),(0,0,1,1,1))
\end{aligned}
$$

$$\mathcal{F}_{15}^{(5)} = ((3,0,0,0,0),(0,2,1,0,0),(0,0,1,2,0),(0,1,0,1,1),(0,0,1,0,2))$$
$$\mathcal{F}_{16}^{(5)} = ((3,0,0,0,0),(0,2,1,0,0),(0,0,1,1,1),(0,1,0,1,1),(0,0,1,1,1))$$
$$\mathcal{F}_{17}^{(5)} = ((2,1,0,0,0),(1,2,0,0,0),(0,0,1,1,1),(0,0,1,1,1),(0,0,1,1,1))$$
$$\mathcal{F}_{18}^{(5)} = ((2,1,0,0,0),(0,2,1,0,0),(1,0,1,1,0),(0,0,0,2,1),(0,0,1,0,2))$$
$$\mathcal{F}_{19}^{(5)} = ((2,1,0,0,0),(0,2,1,0,0),(0,0,1,2,0),(1,0,0,1,1),(0,0,1,0,2))$$
$$\mathcal{F}_{20}^{(5)} = ((2,1,0,0,0),(0,2,1,0,0),(0,0,1,1,1),(1,0,0,1,1),(0,0,1,1,1))$$
$$\mathcal{F}_{21}^{(5)} = ((2,0,1,0,0),(0,2,0,1,0),(1,0,1,1,0),(0,1,0,1,1),(0,0,1,0,2))$$
$$\mathcal{F}_{22}^{(5)} = ((2,0,1,0,0),(0,2,0,1,0),(1,0,1,1,0),(0,0,0,1,2),(0,1,1,0,1))$$
$$\mathcal{F}_{23}^{(5)} = ((2,0,1,0,0),(0,2,0,1,0),(0,0,1,1,1),(1,0,0,1,1),(0,1,1,0,1))$$
$$\mathcal{F}_{24}^{(5)} = ((2,1,0,0,0),(1,1,1,0,0),(0,0,1,1,1),(0,0,1,1,1),(0,1,0,1,1))$$
$$\mathcal{F}_{25}^{(5)} = ((2,1,0,0,0),(0,1,2,0,0),(0,0,1,1,1),(1,0,0,1,1),(0,1,0,1,1))$$
$$\mathcal{F}_{26}^{(5)} = ((2,1,0,0,0),(0,1,1,0,1),(1,0,1,1,0),(0,0,0,2,1),(0,1,1,0,1))$$
$$\mathcal{F}_{27}^{(5)} = ((1,1,1,0,0),(1,1,0,1,0),(0,0,1,1,1),(1,0,0,1,1),(0,1,1,0,1))$$
$$\mathcal{F}_{28}^{(5)} = ((1,1,1,0,0),(1,1,1,0,0),(0,0,1,1,1),(0,1,0,1,1),(1,0,0,1,1))$$

$$n=1: \rho_{\text{ind}} = \pi^{-2}(\,2 - \mathcal{F}_1^{(1)})$$
$$n=2: \rho_{\text{ind}} = \pi^{-3}(4 - 2\mathcal{F}_1^{(1)} + \tfrac{1}{2}\mathcal{F}_1^{(2)} - \tfrac{1}{2}\mathcal{F}_2^{(2)})$$
$$n=3: \rho_{\text{ind}} = \pi^{-4}(12 - 4\mathcal{F}_1^{(1)} + \mathcal{F}_1^{(2)} - \mathcal{F}_2^{(2)} - \tfrac{1}{6}\mathcal{F}_1^{(3)} + \tfrac{1}{2}\mathcal{F}_2^{(3)} - \mathcal{F}_3^{(3)} + \mathcal{F}_4^{(3)} - \tfrac{1}{3}\mathcal{F}_5^{(3)})$$
$$\begin{aligned} n=4: \rho_{\text{ind}} = \pi^{-5}(&48 - 12\mathcal{F}_1^{(1)} + 2\mathcal{F}_1^{(2)} - 2\mathcal{F}_2^{(2)} - \tfrac{1}{3}\mathcal{F}_1^{(3)} + \mathcal{F}_2^{(3)} - 2\mathcal{F}_3^{(3)} + 2\mathcal{F}_4^{(3)} - \tfrac{2}{3}\mathcal{F}_5^{(3)} \\ &+ \tfrac{1}{24}\mathcal{F}_1^{(4)} - \tfrac{1}{4}\mathcal{F}_2^{(4)} + \mathcal{F}_3^{(4)} + \tfrac{1}{8}\mathcal{F}_4^{(4)} - \tfrac{1}{4}\mathcal{F}_5^{(4)} - \mathcal{F}_6^{(4)} - 2\mathcal{F}_7^{(4)} + 2\mathcal{F}_8^{(4)} + \tfrac{1}{3}\mathcal{F}_9^{(4)} + \mathcal{F}_{10}^{(4)} - \mathcal{F}_{11}^{(4)} \end{aligned}$$
$$\begin{aligned} n=5: \rho_{\text{ind}} = \pi^{-6}(&240 - 48\mathcal{F}_1^{(1)} + 6\mathcal{F}_1^{(2)} - 6\mathcal{F}_2^{(2)} - \tfrac{2}{3}\mathcal{F}_1^{(3)} + 2\mathcal{F}_2^{(3)} - 4\mathcal{F}_3^{(3)} + 4\mathcal{F}_4^{(3)} - \tfrac{4}{3}\mathcal{F}_5^{(3)} \\ &+ \tfrac{1}{12}\mathcal{F}_1^{(4)} - \tfrac{1}{2}\mathcal{F}_2^{(4)} + 2\mathcal{F}_3^{(4)} + \tfrac{1}{4}\mathcal{F}_4^{(4)} - \tfrac{1}{2}\mathcal{F}_5^{(4)} - 2\mathcal{F}_6^{(4)} - 4\mathcal{F}_7^{(4)} + 4\mathcal{F}_8^{(4)} + \tfrac{2}{3}\mathcal{F}_9^{(4)} + 2\mathcal{F}_{10}^{(4)} - 2\mathcal{F}_{11}^{(4)} \\ &- \tfrac{1}{120}\mathcal{F}_1^{(5)} + \tfrac{1}{12}\mathcal{F}_2^{(5)} - \tfrac{1}{2}\mathcal{F}_3^{(5)} - \tfrac{1}{8}\mathcal{F}_4^{(5)} + \tfrac{1}{4}\mathcal{F}_5^{(5)} + \tfrac{1}{2}\mathcal{F}_6^{(5)} - \tfrac{3}{5}\mathcal{F}_7^{(5)} + \tfrac{1}{2}\mathcal{F}_8^{(5)} + 2\mathcal{F}_9^{(5)} - 2\mathcal{F}_{10}^{(5)} \\ &- \tfrac{1}{2}\mathcal{F}_{11}^{(5)} + 2\mathcal{F}_{12}^{(5)} + 4\mathcal{F}_{13}^{(5)} - \tfrac{1}{6}\mathcal{F}_{14}^{(5)} - \mathcal{F}_{15}^{(5)} + \mathcal{F}_{16}^{(5)} + \tfrac{1}{6}\mathcal{F}_{17}^{(5)} - 3\mathcal{F}_{18}^{(5)} + \mathcal{F}_{19}^{(5)} \\ &- 3\mathcal{F}_{20}^{(5)} - 4\mathcal{F}_{21}^{(5)} + 2\mathcal{F}_{22}^{(5)} - 2\mathcal{F}_{23}^{(5)} + 2\mathcal{F}_{24}^{(5)} + \mathcal{F}_{25}^{(5)} + \mathcal{F}_{26}^{(5)} + \tfrac{2}{5}\mathcal{F}_{27}^{(5)} - \mathcal{F}_{28}^{(5)}) \end{aligned}$$